

*Google scholar*

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Cooperative Institute for Research in Environmental Sciences (CIRES), University of Colorado Boulder (CU Boulder) and NOAA Global Monitoring Laboratory (GML), 325 Broadway, Boulder, CO 80305

## EDUCATION

**Purdue University**, West Lafayette, IN

Ph.D. in Earth, Atmospheric, and Planetary Sciences

*Aug 2016 – Jul 2020*

Coursework includes Stable Isotope and Forensics, Biogeochemistry, Ecosystem Ecology, Geodata Science, Integrated Global System Modeling, and Terrestrial Ecosystem modeling.

**Princeton University**, Princeton, NJ

M.A. in Atmospheric and Oceanic Sciences

*Aug 2013 – Jul 2015*

Coursework includes Atmospheric Science, Boundary Layer Meteorology, Radiative Transfer, Atmospheric Thermodynamics, Physical Hydrology, Climate-Biosphere Interactions, Geomicrobiology, and Inversion Methods.

**University of California Davis**, Davis, CA

Global Exchange Student

*Sep 2010 – Jun 2011*

**Ewha Womans University**, Seoul, South Korea

B.S. in Environmental Science and Engineering, *Magna cum laude*

*Mar 2009 – Aug 2013*

Double Major in Biological Sciences, Certification in ABEEK Advanced Engineering Program  
Coursework includes Atmospheric Pollution, Water Quality, Environmental Impact Assessment, Mathematical Modeling, Organic Chemistry, Molecular Biology, and Plant ecology.

## RESEARCH

## EXPERIENCE

**CIRES Research Scientist II, CU Boulder and NOAA GML**

*May 2025 – present*

Advancing CarbonTracker-CH<sub>4</sub>, NOAA GML's atmospheric methane assimilation system.

**CIRES Research Scientist I, CU Boulder and NOAA GML**

*Mar 2023 – Apr 2025*

Advancing CarbonTracker-CH<sub>4</sub>, NOAA GML's atmospheric methane assimilation system.

**Postdoctoral Associate, CU Boulder and NOAA GML**

*Sep 2021 – Feb 2023*

Advisor: Dr. Lori Bruhwiler

Joint CH<sub>4</sub> and  $\delta^{13}\text{C}$ -CH<sub>4</sub> inversion modeling of global methane using TM5 atmospheric chemistry model for NOAA's CarbonTracker-CH<sub>4</sub>.

**National Research Council Postdoctoral Fellow, NOAA GML**

*Oct 2020 – Aug 2021*

Advisor: Dr. Lori Bruhwiler

Estimation of CH<sub>4</sub> fluxes and associated isotopic signatures from natural sources and sinks using an improved biogeochemistry model.

**Visiting Scholar, NOAA GML**

*Nov 2018 – Aug 2019*

Advisor: Dr. Edward J. Dlugokencky and Dr. Lori Bruhwiler

Biogeochemistry modeling to understand spatial and temporal variability of carbon isotopic signatures of CH<sub>4</sub> emitted from wetlands.

**NASA Earth and Space Science Fellow, Purdue University**

*Sep 2017 – Jul 2020*

Advisor: Dr. Qianlai Zhuang and Dr. Lisa Welp

Understanding importance of High Affinity Methanotroph in arctic CH<sub>4</sub> budget by using biogeochemistry and atmospheric modeling.

**Graduate Research Assistant, Purdue Isotope Laboratory**

*Jul 2016 – Jul 2020*

Advisor: Dr. Lisa Welp

Seasonal stable carbon isotope variation in temperate deciduous leaves and implications for carbon allocation and water use efficiency estimates using Elemental Analyzer Isotope Ratio Mass Spectrometry.

**Graduate Research Assistant, Princeton University**

Aug 2013 – Apr 2016

Advisor: Dr. David Medvigy

Developing methane microbial dynamics model of Arctic upland soils to elucidate the importance of high affinity methanotrophy.

**Undergraduate Research Assistant, Ewha Womans University**

Sep 2011 – Aug 2013

Advisor: Dr. Yong-Sang Choi

Statistical analysis to study how El Nino-Southern Oscillation and Tropical North Atlantic affect vegetation activity in the Amazon.

**PUBLICATIONS [Under Review]**

Smith, C. C. R., S. Chen, S. L. Malone, G. McNicol, Q. Zhu, L. Liu, and **Oh, Y.** "Integrating machine learning with a process-based model for estimating global wetland methane emissions." *Geophysical Research Letters*, Under Review.

Rozmiarek, K. S., X. Liu, **Oh, Y.**, I. Overeem, E. A. Miller, V. Morris, B. H. Vaughn, K. M. Walter Anthony, C. Edgar, T. Aguirrezabala-Cámpano, B. M. Chase, Z. Tan, N. R. Hasson, Q. Zhuang, C. Smallwood, G. B. Rieker, and T. R. Jones. "Controls on  $^{13}\text{CCH}_4$  from Arctic Thermokarst Lakes." *Journal of Geophysical Research: Biogeosciences*, Under Review.

Qi, Y., Q. Ying, B. Poulter, R. B. Jackson, S. H. Knox, F. Li, **Oh, Y.**, R. Stern, J. D. Watts, S. Wang, L. Xu, X. Zhang, P. Zhao, X. Li, and Z. Zhang. "Critical Challenges in Extending Regional Machine-Learning Models of Wetland  $\text{CH}_4$  Fluxes to the Global Scale." *Global Biogeochemical Cycles*, Under Review.

Sun, Y., R. Yu, R. Dong, S. Chen, L. Liu, **Oh, Y.**, Q. Zhuang, Y. Xie, X. Jia. "Role-Aware Conditional Inference for Spatiotemporal Ecosystem Carbon Flux Prediction." *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (AI for Science Track)*, Under Review.

Wu, Q., K. Gu, B. Peng, Q. Zhu, J. Tang, X. Jia, K. Kumar, M. Chen, C. Lou, **Oh, Y.**, L. Liu. "Constraining global wetland  $\text{CH}_4$  emissions from sparse observations with a Bridge Model framework." *Communications Earth and Environment*, Under Review.

Chen, S., Q. Zhu, L. Liu, J. Zheng, X. Jia, A. Malhotra, S. Metzger, **Oh, Y.**, K. A. Arndt, S. L. Malone, G. McNicol, K. S. Rozmiarek, C. C. R. Smith, F. Yuan, K. Yuan, Y. Ma, Z. Chen, F. Li, E. Ward, Y. Yang, Q. Ying, Q. Zhuang. "Next-Generation Monitoring and Modeling of Wetland Methane Dynamics through Artificial Intelligence." *Communications Earth and Environment*, Under Review.

Watts, J. D., J. Gewirtzman, H. Cadillo-Quiroz, D. Sihi, R. Tang, J. West, V. Leshyk, G. Burba, E. Davidson, M. Helbig, A. Malhotra, J. Matthes, **Oh, Y.**, F.-J. W. Parmentier, B. Poulter, R. K. Varner, X. Xu, H. Zhao, J. Zheng, D. Baldocchi, R. Commane, Z. Dietrich, C. Hettwer, D. Kahn, B. Maglio, D. Monteverde, C. R. Perryman, T. Roychowdhury, B. R. K. Runkle, E. Schuur, R. Vargas, A. Virkkala, Q. Zhuang, and K. Savage. "Terrestrial methane oxidation: Climate implications and research priorities." *Nature Reviews Earth & Environment*, Under Review.

**[Published]**

**Oh, Y.**, L. Bruhwiler, X. Lan, S. Halder, B. Riddell-Young, S. Basu, J. Miller, S. Michel, K. Schuldt, A. Andrews, S. Biraud, Ł. Chmura, T. Di Iorio, E.-A. Guerette, E. Kozlova, L. Liu, S. Montaguti, S. Morimoto, M. Steinbacher, T. Umezawa, I. Xueref-Remy, G. Zazzeri. "Isotopic constraints in methane inversions reveal larger wetland emissions with improved linkage to terrestrial water storage." *Nature Communications*, 2026. <https://doi.org/10.1038/s41467-026-74777-4>

- Oh, Y.,** L. Liu, J. Lee, L. Bruhwiler, X. Lan, S. Michel, S. Basu, J. Miller, Q. Zhu, S. Malone, G. McNicol, Q. Zhuang. "Revising the magnitude and trends of the global methane soil sink with process-based, machine-learning, and atmospheric inversion modeling approaches." *Invited Article on JGR-Biogeosciences*, 2026. <https://doi.org/10.1029/2025JG009668>.
- Zhu, Q., K. A. Arndt, K. Yuan, F. Li, Q. Ying, L. Liu, E. Ward, A. Malhotra, J. Zheng, F. Yuan, S. L. Malone, G. McNicol, S. H. Knox, W. J. Riley, M. S. Torn, S. Chen, B. Riddell-Young, **Oh, Y.,** L. Bruhwiler. "Multi-Decadal Dynamics of Wetland Methane Emissions Revealed by Knowledge-Guided Machine Learning." *Global Change Biology*, 2026. <https://doi.org/10.1111/gcb.70899>
- Dong, R., X. Jia, Y. Sun, L. Liu, **Oh, Y.,** S. Chen. "CHAM-net: A Contrastive Hierarchical Adaptive Meta-network for Robust Global Methane Flux Prediction." *International Joint Conference on Artificial Intelligence*, Accepted.
- Sun, Y., S. Chen, S. Chen, C. Qiu, L. Liu, **Oh, Y.,** S. Malone, G. McNicol, Q. Zhuang, C. C. R. Smith, Y. Xie, X. Jia. "X-MethaneWet: A Cross-scale Global Wetland Methane Emission Benchmark Dataset for Advancing Science Discovery with AI." *KDD 2026 Datasets and Benchmark Track*, 2026. <https://doi.org/10.1145/3770854.3785682>
- Ibarra-Espinosa, S., L. Hu, C. Harkins, B. McDonald, S. Miller, **Oh, Y.,** L. Bruhwiler, C. Sweeney, A. Andrews. "Reduced U.S. Methane Emissions during the COVID-19 Pandemic." *Environmental Science: Atmospheres*, 2026. <https://doi.org/10.1039/d6ea00004e>
- Mirrezaei, M. A., Gaubert, B., Arellano Jr., A. F., Fernandez, R. P., Ortega, I., Emmons, L. K., Kinnison, D., Roozitalab, B., McCain, K., Bruhwiler, L., **Oh, Y.,** Saiz-Lopez, A., Sauniois, M., Feng, C., Xu Y. "Toward Realistic Prognostic Modeling Of The Methane Chemical Loss." *Journal of Geophysical Research - Atmospheres*, 2025. <https://doi.org/10.1029/2025JD045079>
- Riddell-Young, B., S. Michel, X. Lan, P. Tans, T. Röckmann, B. Dasgupta, **Oh, Y.,** L. Bruhwiler, R. Fujita, T. Umezawa, S. Morimoto, J. Miller. "Microbial driver of 2006 – 2023 CH<sub>4</sub> growth indicated by trends in atmospheric  $\delta\text{D-CH}_4$  and  $\delta^{13}\text{C-CH}_4$ ." *Proceedings of the National Academy of Sciences of the United States of America*, 2025. <https://doi.org/10.1073/pnas.2516543122>
- Kuhn, M., Olefeldt, D., Arndt, K. A., Bastviken, D., Bruhwiler, L., Crill, P., DelSontro, T., Fluet-Chouinard, E., Grosse, E., Hovemyr, M., Hugelius, G., MacIntyre, S., Malhotra, A., McGuire, D., **Oh, Y.,** Poulter, B., Treat, C., Turetsky, M., Varner, R., Walter-Anthony, K., Watts, J., Zhang Z. "Current and future methane emissions from Boreal-Arctic wetlands and lakes." *Nature Climate Change*, 2025. doi: 10.1038/s41558-025-02413-y
- Hu, L., Andrews, A., Montzka, S., Miller, S., Bruhwiler, L., **Oh, Y.,** Sweeney, C., McCain, K., Ibarra Espinosa, S., Davis, K., Miles, N., Mountain, M., Lan, X., Crotwell, A., Madronich, M., Mefford, T., Michel, S., Houwelling, S., Miller, J. "An unexpected seasonal cycle in U.S. oil and gas methane emissions." *Environmental Science & Technology*, 2025. doi: 10.1021/acs.est.4c14090
- Oh, Y.,** Bruhwiler, L., Lan, X., Basu, S., Schuldt, K., Thoning, K., Michel, S., et al. "Carbon-Tracker CH<sub>4</sub> 2025." *NOAA Global Monitoring Laboratory*, 2025. doi: 10.25925/hxks-v755.
- Welp, L., **Oh, Y.,** Griffis, T., Yu, Z. "Isotopes of modern atmospheric greenhouse gases: Improving understanding of climate change feedbacks." *Treatise on Geochemistry (Third Edition)* Volume 3, 2025. doi: 10.1016/B978-0-323-99762-1.00125-X.

- Ying, Q., Poulter, B., Watts, J. D., Arndt, K. A., Virkkala, A., Bruhwiler, L., **Oh, Y.**, Rogers, B. M., Natali, S. M., Sullivan, H., Schiferl, L. D., Elder, C., Peltola, O., Bartsch, A., Armstrong, A., Desai, A. R., Euskirchen, E., Göckede, M., Lehner, B., Nilsson, M. B., Peichl, M., Sonnentag, O., Tuittila, E., Sachs, T., Kalhori, A., Ueyama, M., Zhang, Z. "WetCH4: A Machine Learning-based Upscaling of Methane Fluxes of Northern Wetlands during 2016–2022." *Earth System Science Data*, 2024. doi: 10.5194/essd-17-2507-2025
- Lyu, Z., Sommers, P., Schmidt, S. K., Magnani, M., Cimpoiasu, M., Kuras, O., Zhuang, Q., **Oh, Y.**, Fuente, M. D. L., Cramm, M., Bradley, J. A. "Seasonal dynamics of Arctic soils: Capturing year-round processes in measurements and soil biogeochemical models." *Earth-Science Reviews* Volume 254, 2024. doi: 10.1016/j.earscirev.2024.104820.
- Treat, C., Virkkala, A., Burke, E., Bruhwiler, L., Chatterjee, A., Fisher, J., Hashemi, J., Parmentier, F., Rogers, B.M., Westermann, S., Watts, J., Blanc-Betes, E., **Oh, Y.**, Fuchs, M., Kruse, S., Malhotra, A., Miner, K., Schuur, E., Hugelius, G. "Permafrost carbon: progress understanding controls, stocks, and fluxes across terrestrial ecosystems." *Invited review for JGR-Biogeosciences*, 2024. doi: 10.1029/2023JG007638
- Oh, Y.**, Bruhwiler, L., Lan, X., Basu, S., Schuldt, K., Thoning, K., Michel, S., et al. "Carbon-Tracker CH<sub>4</sub> 2023." *NOAA Global Monitoring Laboratory*, 2023. doi: 10.25925/40JT-QD67.
- D'Imperio, L., Li, B., Tiedje, J.M., **Oh, Y.**, Christiansen, J.R., Kepfer-Rojas, S., Westergaard-Nielsen, A., Brandt, K.K., Holm, P.E., Wang, P., Ambus, P., and Elberling, B. "Spatial controls of potential methane oxidation in upland top soils across climatic and geological regions in Greenland." *Nature Communications Earth and Environment*, 2023. doi:10.1038/s43247-023-01143-3.
- Lee, J., **Oh, Y.**, Lee, S., Seo, Y., Yun, J., Yang, Y., Kim, J., Zhuang, Q. and Kang, H. "Soil organic carbon is a key determinant of CH<sub>4</sub> sink in global forest soils." *Nature Communications*, 2023. doi: 10.1038/s41467-023-38905-8.
- Nisbet, E.G., Manning, M.R., Dlugokencky, E.J., Michel, S.E., Lan, X., Roeckmann, T., Gon, H.A.D.V.D., Palmer, P., **Oh, Y.**, Fisher, R. and Lowry, D. "Atmospheric methane: Comparison between methane's record in 2006-2022 and during glacial terminations." *Global Biogeochemical Cycles*, 2023. doi: 10.1029/2023GB007875.
- Zhou, L., Warner, J., Nalli, N.R., Wei, Z., **Oh, Y.**, Bruhwiler, L., Liu, X., Divakarla, M., Pryor, K., Kalluri, S. and Goldberg, M.D. "Spatiotemporal Variability of Global Atmospheric Methane Observed from Two Decades of Satellite Hyperspectral Infrared Sounders." *Remote Sensing*, 2023. doi: 10.3390/rs15122992.
- Zhuang, Q., Guo, M., Melack, J.M., Lan, X., Tan, Z., **Oh, Y.**, and Leung, L.R. "Current and future global lake methane emissions: A process-based modeling analysis." *JGR Biogeosciences*, 2023. doi.org/10.1029/2022JG007137.
- Oh, Y.**, Zhuang, Q., Welp, L., Liu, L., Lan, X., Basu, S., Dlugokencky, E., Bruhwiler, L., Miller, J., Michel, S., Schwietzke, S., Tans, P., Ciais, P., and Chanton, J. "Improved global wetland carbon isotopic signatures support post-2006 microbial methane emission increase." *Nature Communications Earth and Environment*, 2022. doi: 10.1038/s43247-022-00488-5.
- Basu, S., Lan, X., Dlugokencky, E., Bruhwiler, L., **Oh, Y.** "Estimating sources of methane consistent with atmospheric measurements of CH<sub>4</sub> and  $\delta^{13}\text{C}$ -CH<sub>4</sub>." *Atmospheric Chemistry and Physics Discussions*, 2022. doi: 10.5194/acp-2022-317.
- Malone, S., **Oh, Y.**, Arndt, K., Burba, G., Commane, R., Contosta, A., Goodrich, J., Loescher, H., Starr, G., Varner, R. "Gaps in Network Infrastructure limit our understanding of biogenic methane emissions in the United States." *Biogeosciences*, 2021. doi: 10.5194/bg-2021-256.

Lan, X., Basu, S., Schwietzke, S., Bruhwiler, L.M.P., Dlugokencky, E.J., Michel, S.E., Sherwood, O.A., Tans, P.P., Thoning, K., Etiope, G. and Zhuang, Q., Liu, L., **Oh, Y.**, Miller, J. B., Petron, G., Vaughn, B. H., and Crippa, M. “Improved Constraints on Global Methane Emissions and Sinks Using  $\delta^{13}\text{C-CH}_4$ .” *Global Biogeochemical Cycles*, 2021. doi: 10.1029/2021GB007000.

**Oh, Y.**, Zhuang, Q., Liu, L., Welp, L., Lau, M., Onstott, T., Medvigy, D., Bruhwiler, L., Dlugokencky, D., Hugelius, G., D’Imperio, L., and Elberling, B. “Reduced net methane emissions due to microbial methane oxidation in a warmer Arctic.” *Nature Climate Change*, 2020. doi: 10.1038/s41558-020-0734-z.

Abbasi, A. O., A. Salazar, **Oh, Y.**, Reinsch, S., Uribe, M., Li, J., Rashid, I., and Dukes, J. “Soil responses to manipulated precipitation changes: A synthesis of meta-analyses.” *Biogeosciences*, 2020. doi: 10.5194/bg-17-3859-2020.

Liu, L., Zhuang, Q., **Oh, Y.**, Shurpali, N.J., Kim, S. and Poulter, B. “Uncertainty Quantification of Global Net Methane Emissions from Terrestrial Ecosystems Using a Mechanistically-based Biogeochemistry Model.” *Journal of Geophysical Research: Biogeosciences*, 2020. doi: 10.1029/2019JG005428.

Lau, M., Harris, R., **Oh, Y.**, Yi, M., Behmard, A., and Onstott, T. “Taxonomic and functional compositions impacted by the quality of metatranscriptomic assemblies.” *Frontiers in Microbiology*, 2018. doi:10.3389/fmicb.2018.01235.

**Oh, Y.**, Stackhouse, B., Lau, M., Xu, X., Trugman, A., Moch, J., Onstott, T., Jørgensen, C., D’Imperio, L., Elberling, B., Emmerton, C., St Louis, V., and Medvigy, D. “A scalable model for methane consumption in arctic mineral soils.” *Geophysical Research Letters*, 2016. doi:10.1002/2016GL069049.

Lau, M., Kieft, T., Kuloyo, O., Linage, B., Heerden, E., Lindsay, M., Magnabosco, C., Wang, W., Wiggins, J., Guo, L., Perlman, D., Kyin, S., Shwe, H., Harris, R., **Oh, Y.**, Yi, M., Purtschert, R., Slater, G., Ono, S., Wei, S., Li, L., Lollar, B., and Onstott, T. “An oligotrophic deep-subsurface community dependent on syntrophy is dominated by sulfur-driven autotrophic denitrifiers.” *Proceedings of the National Academy of Sciences*, 2016. doi: 10.1073/pnas.1612244113.

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| TEACHING<br>&<br>MENTORING | <b>Research Mentor, NOAA GML and CU Boulder</b> <span style="float: right;"><i>Mar 2025 – present</i></span><br>Postdoc: Santanu Halder (NOAA GML and CU Boulder)<br>Incorporating Deuterium isotope into CarbonTracker-CH <sub>4</sub> .                                                     |
|                            | <b>Research Mentor, NOAA GML and CU Boulder</b> <span style="float: right;"><i>Sep 2025 – present</i></span><br>Students: Rongchao Dong, Yiming Sun (University of Pittsburg)<br>Knowledge-guided machine learning estimation of global methane wetland emissions and soil sinks.             |
|                            | <b>CIRES Mentor, NOAA GML and CU Boulder</b> <span style="float: right;"><i>Oct 2023 – present</i></span><br>Mentees: Frank Smith (2023) and Gonzalo Ferrada (2024) (NOAA GML and CU Boulder)<br>Help mentees on professional growth and career development.                                  |
|                            | <b>Research Mentor and Ph.D. Committee, NOAA GML and CU Boulder</b> <span style="float: right;"><i>Sep 2022 – present</i></span><br>Student: Kevin Rozmiarek, Ph.D. student at CU Boulder<br>Biogeochemistry modeling of Arctic wetland isotope signatures.                                   |
|                            | <b>CIRES Advisor, NOAA GML and CU Boulder</b> <span style="float: right;"><i>Mar 2024 – Jul 2025</i></span><br>Postdoc: Chris C. Smith (NOAA GML and CU Boulder, now Assistant Professor at Indiana University)<br>Knowledge-guided machine learning estimation of global methane soil sinks. |

**Research Mentor and Advisor, NOAA GML and CU Boulder** *Sep 2022 – Aug 2024*

Student: Matt Ziminski, Master student at University of Illinois Chicago  
4-D visualization of CarbonTracker-CH<sub>4</sub>.

**Guest Lecture, Department of Geography, University of Colorado Boulder** *Apr 2026*

Course: GEOG 5100-002 Special Topic: Carbon Cycle

Next-Generation Modeling of Global Natural Methane Fluxes - Integrating Multi-scale Observations with Knowledge-Guided Machine Learning.

**Guest Lecture, School of the Environment, Yale University** *Sep 2023*

Course: ENV 623 - The role of Methane in Global Climate Disruption

Recent Atmospheric Methane Growth, Global Methane Pledge, and the CarbonTracker-CH<sub>4</sub>.

**Guest Lecture, Earth, Atmospheric, and Planetary Sciences, Purdue University** *Oct 2019*

Course: EAPS 591 – Stable Isotopes

Global methane cycle and the stable isotopes.

**Guest Lecture, Earth, Atmospheric, and Planetary Sciences, Purdue University** *Oct 2019*

Course: EAPS 529 : Modeling Ecosystems and Biogeochemical Cycles

Microbial methane oxidation projects reduced net methane emissions in a warmer Arctic.

**Research Mentor, Princeton University**

*Sep 2016 – May 2017*

Student: Christianese Kaiser, Senior Student at Princeton University

Estimating annual CH<sub>4</sub> sinks in the Arctic of the 2000s using regional-level methane microbial dynamics model.

GRANT &

FELLOWSHIP

**Co-Investigator**, National Science Foundation

*pending*

Principal Investigator: Ruth Varner and Kelly Wrighton

*Total \$30,000,000*

The WeM5 Science and Technology Center (STC): Building A Unified framework for Wetland Monitoring, Modeling, and Management of Methane and Microbes.

**Principal Investigator**, Google Impact Challenge: AI for Science

*pending*

Co-Is: Licheng Liu

*Total \$700,000*

AI-Enabled Earth System Digital Twin for Methane Variability and Predictability.

**Co-Investigator**, Heising-Simons Foundation

*pending*

Principal Investigator: : Licheng Liu

*Total \$65,000*

The 2028 AI4Methane Workshop: Bridging Methane Science and AI with Knowledge-Guided Machine Learning and Harmonized Datasets.

**Principal Investigator**, Spark Climate Solutions

*Nov 2025*

Co-PIs: L. Liu, S. Malone, and G. McNicol

*Total \$40,000*

AI for Natural Methane Working Group: Advancing Methane Science with Data Harmonization and Knowledge-Guided Machine Learning.

**Principal Investigator**, NSF Environmental Data Science Innovation/Inclusion Lab *Jan 2024*

Co-PIs: S. Malone, G. McNicol, and L. Liu *Total \$65,000 + Logistic/Computing Resources*

AI for Natural Methane Working Group: Harmonizing Natural Methane Datasets using Knowledge Guided Machine Learning.

**Principal Investigator**, Department of Energy

*Aug 2023*

Collaborator: L. Liu, S. Malone, Q. Zhu, G. McNicol, and Z. Jin

*Total \$400,000*

Estimation of global methane soil sink using synthesized datasets and knowledge-guided machine learning.

**Collaborator**, National Oceanic and Atmospheric Administration (NOAA)

*June 2023*

Principal Investigator: Lori Bruhwiler and Daniel Jacobs

*Total \$700,000*

Advancing NOAA's CarbonTracker-CH<sub>4</sub> Budget.

**Collaborator**, National Oceanic and Atmospheric Administration (NOAA) *June 2023*  
Principal Investigator: Benjamin Gaubert *Total \$700,000*  
Role of atmospheric chemistry in monitoring changes in the CH<sub>4</sub> Budget.

**Collaborator**, Department of Defense *May 2023*  
Principal Investigator: Merritt Turetsky *Total \$6,250,000*  
Synoptic measurement of stream and atmospheric indicators to improve the monitoring and prediction of climate-induced permafrost degradation across Alaska.

**Principal Investigator**, National Research Council Postdoctoral Fellowship *Oct, 2020*  
Joint CH<sub>4</sub> and  $\delta^{13}\text{C}$ -CH<sub>4</sub> inversion modeling of global methane budget. *Total \$60,000.*

**Principal Investigator**, NASA Earth and Space Science Fellowship *Sep 2017*  
B-U and T-D modeling of Arctic CH<sub>4</sub> budget by considering novel microbes. *Total \$150,000.*

HONORS & AWARDS

Biogeosciences Leaders of Tomorrow, AGU Biogeosciences *May 2025*

Global Monitoring Annual Conference Early Career Award, NOAA GML *May 2022*

CESM Polar Modeling Workshop Scholarship, NCAR, Boulder, CO *Aug 2018*

National Research Council Postdoctoral Fellowship *Oct 2020 – Aug 2021*

NASA Earth and Space Science Fellowship *Sep 2017 – Aug 2020*

Henry Silver Graduate Scholarship, Purdue University *May 2018*

Ross Fellowship in Natural Sciences and Engineering, Purdue University *Aug 2016 – Jul 2017*

Purdue Climate Change Research Center Award *Aug 2016*

Research Fellowship in Natural Science and Engineering, Princeton University *Sep 2013 – Feb 2016*

SELECTED PRESENTATION

*Invited, June 2026.* Earth Lab - Environmental Data Science Seminar Series, Boulder, CO.

*Invited, May 2026.* ESHL Innovation Summit 2026, Boulder, CO.

*Contributed, May 2026.* NOAA GML 53 rd Global Monitoring Annual Conference, Boulder, CO.

*Invited, Mar 2026.* Iowa State University - Department of the Earth, Atmosphere, and Climate, Ames, IA.

*Contributed (2 talks), Dec 2025.* American Geophysical Union Fall Meeting, San Francisco, CA.

*Invited, Nov 2025.* Cornell University - Department of Ecology and Evolutionary Biology, Ithaca, New York.

*Invited, Nov 2025.* Colorado State University - Department of Atmospheric Sciences, Fort Collins, Colorado.

*Invited, Nov 2025.* Workshop on Uniting Modelers, Microbiologists, and Biogeochemists to Improve Wetland Methane Prediction, Columbus, Ohio.

*Invited, Oct 2025.* NCAR Mesoscale & Microscale Meteorology Group, Boulder, Colorado.

*Invited, May 2025.* Yale AI for Natural Carbon Capture Workshop, New Haven, Connecticut.

*Invited, Apr 2025.* U.S. Department of Energy - Environmental System Science Program PI Meeting, Reston, Virginia.

*Invited, Jan 2025.* National University Singapore - Department of Civil and Environmental Engineering, Singapore.

*Contributed, Dec 2024.* American Geophysical Union Fall Meeting, San Francisco, CA.

*Invited, Aug 2024.* Woodwell Climate Research Center Methane Workshop, Orono, ME.

*Invited, May 2024.* California Institute of Technology - Department of Environmental Science and Engineering, Pasadena, CA.

*Contributed, Apr 2024.* DOE Environmental System Science (ESS) PI Meeting, Reston, VA.

*Invited, Mar 2024.* Spark Climate Solutions Brown Bag Seminar Series, Boulder, CO.

*Invited, Mar 2024.* NOAA AI Working Group Meeting, Boulder, CO.

*Invited, Feb 2024.* Yale University - Department of Earth & Planetary Sciences, New Haven, CT.

*Invited, Feb 2024.* University of Texas - Department of Sustainable Earth Systems Sciences, Dallas, TX.

*Invited, Jan 2024.* Earth Lab Environmental Data Science Seminar Series, Boulder, CO.

*Contributed, Dec 2023.* American Geophysical Union Fall Meeting, San Francisco, CA.

*Invited, Nov 2023.* NASA Jet Propulsion Laboratory Carbon Club Seminar, Pasadena, CA.

*Invited, Oct 2023.* Atmoschem GitHub Seminar, São Paulo, Brazil.

*Invited, Sep 2023.* Yale Observation Infrastructure for Natural Methane Emissions Workshop, New Haven, CT.

*Contributed, Jul 2023.* NOAA National Environmental Satellite, Data, and Information Service (NESDIS) Workshop, Washington, D.C.

*Invited, Jul 2023.* Seoul National University - Department of Landscape Architecture and Rural Systems Engineering, Seoul, Korea.

*Invited, Jul 2023.* Seoul National University - Department of Environmental Planning / Environmental Management, Seoul, Korea.

*Invited, Jul 2023.* Yonsei University - Department of Civil and Environmental Engineering, Seoul, Korea.

*Invited, Apr 2023.* NOAA-NASA Joint Polar Satellite System Science Seminar Series, Washington, D.C.

*Invited, Apr 2023.* DOE Artificial intelligence for methane (AI4CH<sub>4</sub>) Workshop, Richland, WA.

*Invited, Mar 2023.* DOE Lawrence Berkeley National Laboratory - Climate & Ecosystem Sciences Division Seminar, New Haven, CT.

*Invited, Dec 2022.* American Geophysical Union Fall Meeting, Chicago, IL.

*Invited, Oct 2022.* University of Illinois Chicago - Department of Earth and Environmental Sciences, Chicago, IL.

*Contributed, Sep 2022.* DOE Ameriflux Annual Meeting 2022, Pellston, MI.

*Contributed, Aug 2022.* WMO - BIPM Workshop on Metrology for Climate Action, Paris, France.

*Contributed, May 2022.* NOAA GML 50st Global Monitoring Annual Conference, Boulder CO.

*Invited, Apr 2022.* CU Boulder - Institute of Arctic and Alpine Research, Boulder, CO.

*Contributed, Dec 2021.* American Geophysical Union Fall Meeting, New Orleans, LA.  
*Invited, Oct 2021.* The 10th Asia-Pacific GAW Workshop on GHG, Seoul, South Korea.  
*Invited, Sep 2021.* Ulsan National Institute of Science & Technology, Ulsan, South Korea.  
*Invited, May 2020.* NOAA GML Global Monitoring Annual Conference, Boulder CO.  
*Invited, Oct 2019.* Seminar at the University of Copenhagen - Center for Permafrost, Copenhagen, Denmark.  
*Invited, Jun 2016.* Korea Polar Research Institute Workshop, Incheon, South Korea.  
*Invited, May 2016.* South University of Science and Technology Workshop, Shenzhen, China

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                            |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| COMMUNITY SERVICES | <b>Principal Investigator, AI for Natural Methane Working Group</b>                                                                                                                                                                                                                                                                                                                                                                                                                  | <i>Mar 2024 – present</i>  |
|                    | Working Group for Harmonizing Natural Methane Datasets using Knowledge Guided Machine Learning.                                                                                                                                                                                                                                                                                                                                                                                      |                            |
|                    | <b>Organizing Committee, NOAA GML Seminar Series</b>                                                                                                                                                                                                                                                                                                                                                                                                                                 | <i>Sep 2024 – present</i>  |
|                    | Planning and Hosting Monthly NOAA GML Seminar Series.                                                                                                                                                                                                                                                                                                                                                                                                                                |                            |
|                    | <b>Organizing Committee, CIRES mentoring program, CU Boulder</b>                                                                                                                                                                                                                                                                                                                                                                                                                     | <i>Jul 2023 – present</i>  |
|                    | Mentorship program where CIRES mentor and mentee meet every month to foster professional growth and career development.                                                                                                                                                                                                                                                                                                                                                              |                            |
|                    | <b>Early Career Scientist Committee, FLUXNET Network</b>                                                                                                                                                                                                                                                                                                                                                                                                                             | <i>Aug 2022 – present</i>  |
|                    | The FLUXNET Early Career Network hosts seminars and provides a platform regarding research, career, and funding opportunities for early-career scientists.                                                                                                                                                                                                                                                                                                                           |                            |
|                    | <b>Session Organizer, American Geophysical Union Fall Meeting</b>                                                                                                                                                                                                                                                                                                                                                                                                                    | <i>Dec 2018 – present</i>  |
|                    | Organizing sessions about Terrestrial and Aquatic Methane Sinks in 2018, Data-driven Estimation of Global Wetland Fluxes in 2023 and 2024, and Verification on Anthropogenic Emissions in 2025-2026.                                                                                                                                                                                                                                                                                 |                            |
|                    | <b>Education Outreach Volunteer, Earth Science Program, Purdue University</b>                                                                                                                                                                                                                                                                                                                                                                                                        | <i>Aug 2016 – Aug 2020</i> |
|                    | Outreach program to local elementary school where Purdue graduates host scientific programs.                                                                                                                                                                                                                                                                                                                                                                                         |                            |
|                    | <b>Member, Women in GeoScience Program, Princeton University</b>                                                                                                                                                                                                                                                                                                                                                                                                                     | <i>Sep 2014 – Aug 2016</i> |
|                    | Mentorship program where Princeton graduate students meet senior scientists every week to get advice on successful career as women scientists.                                                                                                                                                                                                                                                                                                                                       |                            |
|                    | <b>Peer Review for more than 30 papers</b>                                                                                                                                                                                                                                                                                                                                                                                                                                           | <i>2020 – present</i>      |
| ET CETERA          | Journals including <i>Nature Climate Change</i> , <i>Nature Communications Earth and Environment</i> , <i>Science Advances</i> , <i>Journal of Climate</i> , <i>Nature npj Climate and Atmospheric Science</i> , <i>National Science Review</i> , <i>Carbon Balance and Management</i> , <i>Geophysical Research Letters</i> , <i>Global Change Biology</i> , <i>Proceedings of the National Academy of Sciences of the United States of America</i> , and <i>Waste Management</i> . |                            |
|                    | <b>Modeling Skills:</b> Process-based Terrestrial Ecosystem Model (TEM), TM5 Atmospheric Chemistry and Transport Model, and other Earth System Models (CESM, ORCHIDEE, LPJ-WhyMe). AI-based Knowledge-Guided Machine Learning (KGML). Analyzing the model output with Matlab, Python, Fortran, IDL, C, C++, L <sup>A</sup> T <sub>E</sub> X(experienced)                                                                                                                             |                            |
|                    | <b>Laboratory Skills:</b> Gas chromatography/isotope ratio mass spectrometry (GC/IRMS)                                                                                                                                                                                                                                                                                                                                                                                               |                            |
|                    | <b>Language:</b> English (Fluent), Korean (Fluent), Chinese (Beginner), Japanese (Beginner)                                                                                                                                                                                                                                                                                                                                                                                          |                            |

**Updated in July 2026**